DATE: EXPT NO:04	DATA VISUALIZATION

AIM:

To visualize the Earthquake dataset using Python and create different charts to understand earthquake patterns, magnitude distributions, and identify important geographic features.

PROCEDURE:

1. Start the program.
2. Import Pandas, Matplotlib, and Seaborn libraries.
3. Load the Earthquake CSV dataset.
4. Display the first few records using head().
5. Create a bar chart to compare seismic events based on magnitude.
6. Create a box plot to show the distribution of earthquake magnitudes.
7. Create a line chart to analyze seismic trends across recorded events.
8. Add suitable titles, axis labels, and legends.
9. Display all the visualizations.
10. Analyze the graphs and identify important patterns.
11. Stop the program.

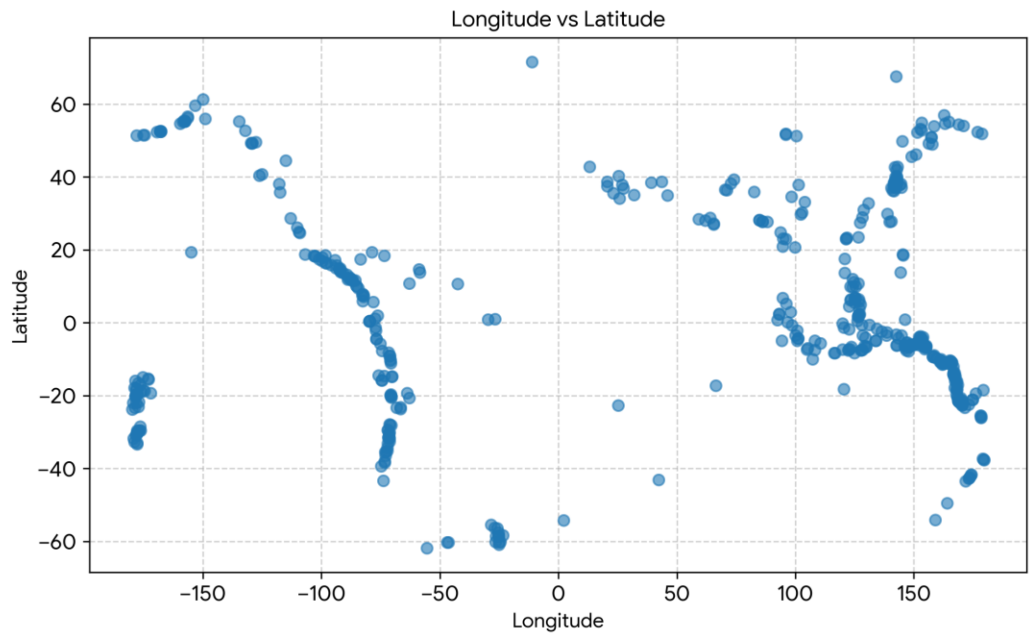
CODE:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Load dataset
df = pd.read_csv("earthquake_data.csv")

# Select numerical columns
num_cols = df.select_dtypes(include="number").columns
print("Numerical columns:", num_cols.tolist())

# BAR CHART
plt.figure(figsize=(8, 5))
df[num_cols[0]].value_counts().head(10).plot(kind="bar")
plt.title("Bar Chart")
plt.xlabel(num_cols[0])
plt.ylabel("Count")
plt.show()

# BOX PLOT
```

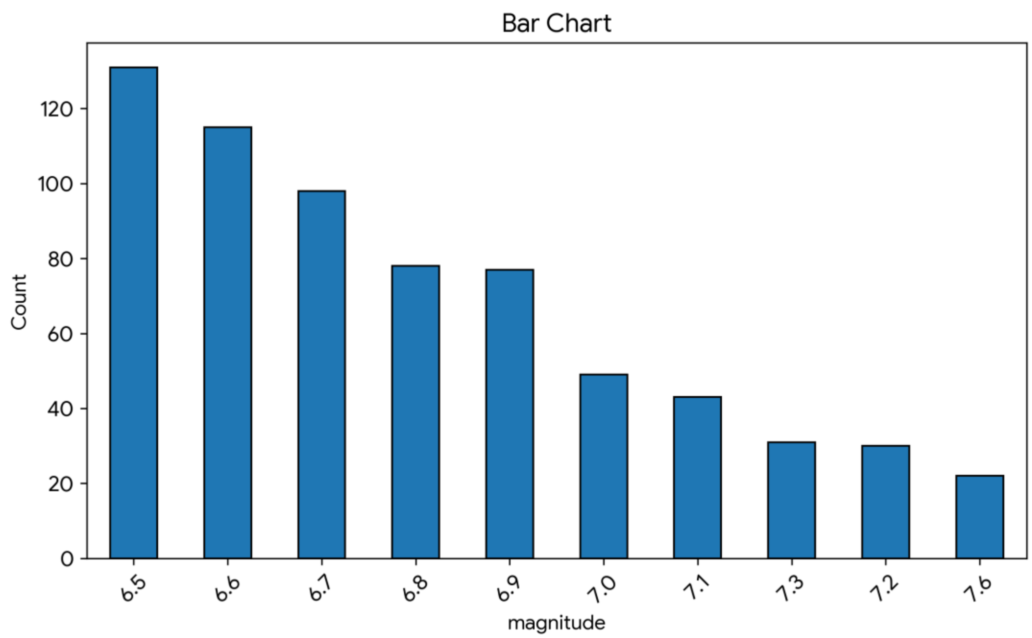
```
plt.figure(figsize=(8, 5))
sns.boxplot(x=df[num_cols[0]])
plt.title("Box Plot")
plt.xlabel(num_cols[0])
plt.show()

# LINE CHART
plt.figure(figsize=(8, 5))
plt.plot(df[num_cols[0]].head(20), marker="o")
plt.title("Line Chart")
plt.xlabel("Earthquake Event")
plt.ylabel(num_cols[0])
plt.grid()
plt.show()
```

OUTPUT:

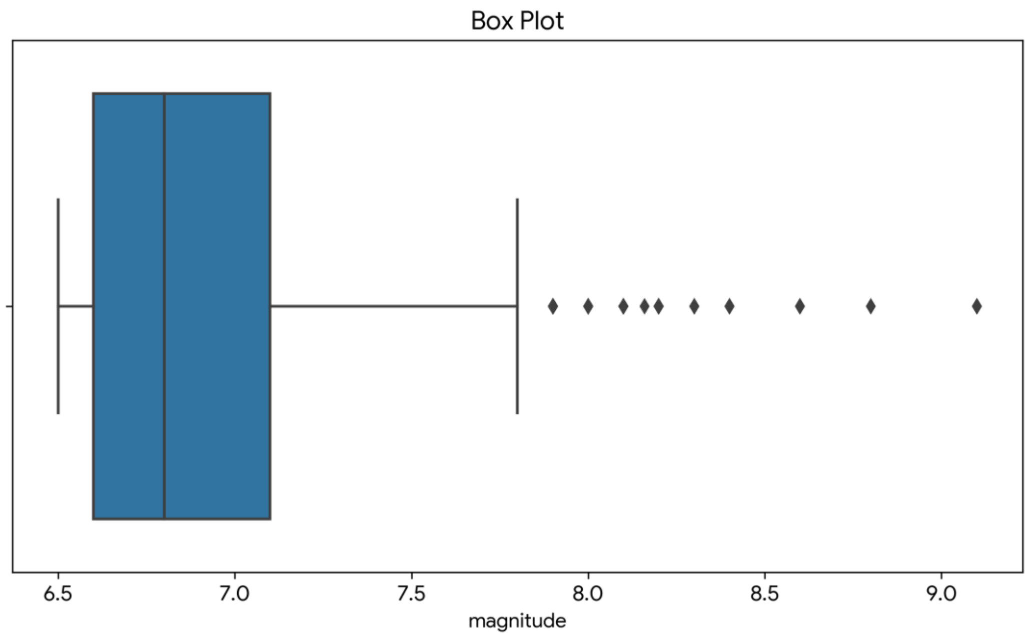
```
Columns:
['title', 'magnitude', 'date_time', 'cdi', 'mmi', 'alert', 'tsunami', 'sig',
'net', 'nst', 'dmin', 'gap', 'magType', 'depth', 'latitude', 'longitude',
'location', 'continent', 'country']

Numerical columns: ['magnitude', 'cdi', 'mmi', 'tsunami', 'sig', 'nst', 'dmin',
'gap', 'depth', 'latitude', 'longitude']

First 5 Records:
                                          title  magnitude         date_time  cdi
mmi  alert  tsunami  sig net  nst   dmin   gap magType    depth  latitude
longitude                  location continent          country
0  M 7.0 - 18 km SW of Malango, Solomon Islands        7.0  22-11-2022 02:03    8
7  green        1  768  us  117  0.509  17.0     mww   14.000   -9.7963    159.596
Malango, Solomon Islands   Oceania  Solomon Islands
1      M 6.9 - 204 km SW of Bengkulu, Indonesia        6.9  18-11-2022 13:37    4
4  green        0  735  us   99  2.229  34.0     mww   25.000   -4.9559    100.738
Bengkulu, Indonesia       NaN             NaN
2                                       M 7.0 -        7.0  12-11-2022 07:09    3
3  green        1  755  us  147  3.125  18.0     mww  579.000  -20.0508   -178.346
NaN   Oceania             Fiji
3           M 7.3 - 205 km ESE of Neiafu, Tonga        7.3  11-11-2022 10:48    5
5  green        1  833  us  149  1.865  21.0     mww   37.000  -19.2918   -172.129
Neiafu, Tonga       NaN             NaN
4                                       M 6.6 -        6.6  09-11-2022 10:14    0
2  green        1  670  us  131  4.998  27.0     mww  624.464  -25.5948    178.278
NaN       NaN             NaN
```

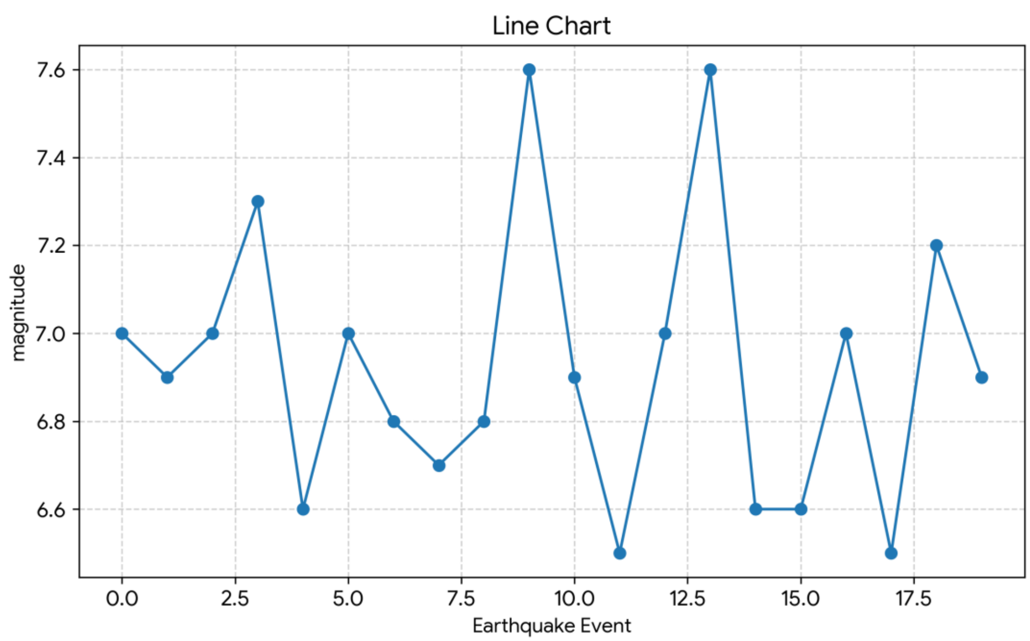


Longitude vs Latitude
Latitude
Longitude
60
40
20
0
−20
−40
−60
−150
−100
−50
0
50
100
150



Bar Chart
Count
magnitude
120
100
80
60
40
20
0
6.5
6.6
6.7
6.8
6.9
7.0
7.1
7.3
7.2
7.6





RESULT:

The Earthquake dataset was successfully imported, analyzed, and visualized using various charts including bar charts, box plots, scatter plots, and line charts. Important patterns regarding seismic magnitudes, geographical distribution, and depth characteristics were identified through data visualization.